



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
FULL SYLLABUS
08-04-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE (ALL STAR BATCH)]

PAPER-2

Time : 3 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **36** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

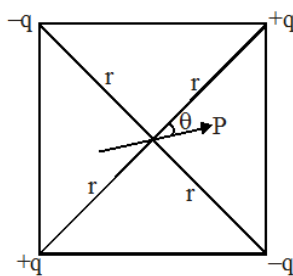
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

- Given a square frame of diagonal length $2r$ made of insulating wires. There is a short dipole, having dipole moment P , fixed in the plane of the figure lying at the center of the square, making an angle θ as shown in figure. Four identical particles having charges of magnitude q each and alternatively positive and negative sign are placed at the four corners of the square. Select the correct alternative(s).



- Electrostatic force on the system of four charges due to dipole is $\frac{6kPq}{r^3}$.
- Electrostatics force on the system of four charges due to dipole is $\frac{6kPq}{r^3} \cos \theta$.
- Net torque on the system of four charges about the centre of the square due to dipole is zero.
- Net torque on the system of four charges about the centre of the square due to dipole is $\frac{3kPq}{r^2}$.

Space for Rough Work

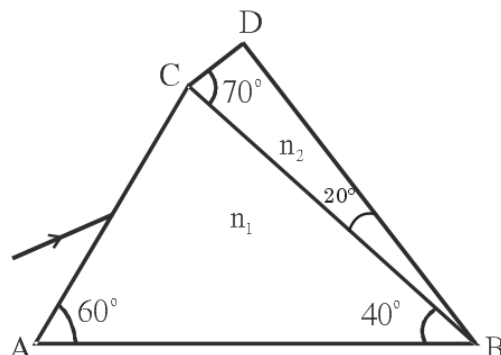
2. A wooden cross floats in water. Each arm of the cross is coated on one side with varnish as shown in the figure. Then mark the correct statement(s).



- (A) The water will rise to different heights on the both sides of each arm.
 (B) The wetting angle on the both sides of each arm will be different.
 (C) The cross will rotate because of force due to surface tension.
 (D) The cross will not rotate because of force due to surface tension.
3. The kinetic energy of α -particle which flies out of the nucleus of radium atom ${}^{226}_{88}\text{Ra}$ in radioactive disintegration is 4.78 MeV. Then mark the correct statement(s)
- (A) The speed of the α -particle is 1.55×10^6 m/s.
 (B) The speed of the α -particle is 2.39×10^7 m/s.
 (C) The total energy evolved during the escape of the α -particle is 8.87 MeV.
 (D) The total energy evolved during the escape of the α -particle is nearly 4.87 MeV.

Space for Rough Work

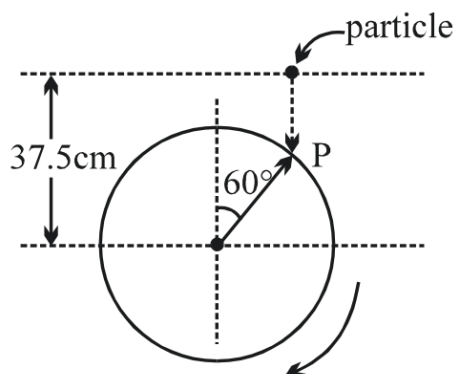
4. A prism of refractive index n_1 and another prism of refractive index n_2 are stuck together without a gap as shown in the figure. The angles of the prisms are as shown. n_1 and n_2 depend on λ , the wavelength of light according to $n_1 = 1.20 + \frac{10.8 \times 10^4}{\lambda^2}$ & $n_2 = 1.45 + \frac{1.80 \times 10^4}{\lambda^2}$ where λ is in nm. Then mark the correct statement(s)



- (A) The wavelength λ_0 for which rays incident at any angle on the interface BC pass through without bending at that interface is 6×10^2 nm.
- (B) The corresponding value of n_2 for $\lambda = \lambda_0$ will be 1.5.
- (C) The corresponding value of n_1 for $\lambda = \lambda_0$ will be 1.6.
- (D) The angle of incidence 'i' on the face AC for $\lambda = \lambda_0$ such that the deviation produced by the combination of prisms is minimum is 60°.

Space for Rough Work

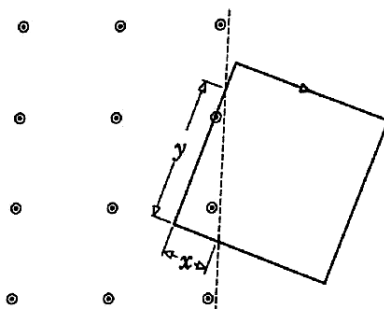
5. A uniform solid wheel of mass m and radius 15 cm is free to rotate without friction about a fixed horizontal axis passing through its centre. A particle of equal mass m strikes it at a point P after falling vertically downward as shown in figure, and sticks to the wheel. Then mark the correct option(s).



- (A) The maximum angular speed of the wheel is $\omega_f = \frac{20}{3}\sqrt{5}\text{ rad/s}$.
- (B) The magnitude of force acting on the wheel when its angular speed becomes maximum is $\frac{16\text{ mg}}{3}$.
- (C) The initial angular speed of the wheel is $\frac{20\sqrt{2}}{3}\text{ rad/s}$.
- (D) The magnitude of force acting on the wheel when its angular speed becomes maximum is $\frac{8\text{ mg}}{3}$.

Space for Rough Work

6. A rigid square loop of side length ' ℓ ' carrying an electric current is held motionless on a frictionless horizontal tabletop. A uniform magnetic field pointing upwards is switched on everywhere to the left of the dashed line as shown in the figure and then the loop is released. Considering different lengths of the side segments x and y ($x < y$), which of the following conclusions can you make?

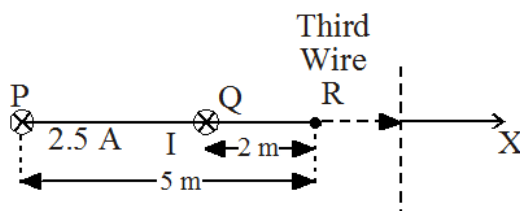


- (A) If $x < \frac{\ell}{2}$ and $y = \frac{\ell}{2}$, the loop starts rotating anticlockwise.
- (B) If $x < \frac{\ell}{2}$ and $y < \frac{\ell}{2}$, the loop starts rotating anticlockwise.
- (C) If $x < \frac{\ell}{2}$ and $y > \frac{\ell}{2}$, the loop starts rotating clockwise.
- (D) If $x < \frac{\ell}{2}$ and $y > \frac{\ell}{2}$, more information is required to decide which way the loop starts rotating.
7. Two identical heaters are coated with paint. In 1st case $e_1 = 1.0$ and in 2nd case $e_2 = 0.5$. Both are kept in identical chambers which are in similar surroundings. If the heaters are switched on, in steady state 1st heater has temperature T_1 on surface and θ_1 of its chamber. 2nd heater has temperature T_2 on surface and θ_2 of its chamber. Then select the correct option(s).

- (A) $\theta_1 = \theta_2$ (B) $T_1 < T_2$ (C) $\theta_1 > \theta_2$ (D) $T_1 = T_2$

Space for Rough Work

8. Two long parallel wires carrying current 2.5 ampere and I ampere in the same direction (directed into the plane of the paper) are held at P and Q respectively such that they are perpendicular to the plane of the paper. The points P and Q are located at a distance of 5 metre and 2 metre respectively from a collinear point R.



- (A) If an electron moving with a velocity of 4×10^5 m/s along the positive X-direction experiences a force of magnitude 3.2×10^{-20} N at the point R then the value of current I is 4A.
- (B) Using the value of I from option (A), if a third long parallel wire carrying a current of magnitude 2.5 ampere may be placed to the right of point R by 1m along x-axis then the magnetic induction at R will be zero.
- (C) If an electron moving with a velocity of 4×10^5 m/s along the positive X-direction experiences a force of magnitude 3.2×10^{-20} N at the point R then the value of current I is 8A.
- (D) Using the value of I from option (A), if a third long parallel wire carrying a current of magnitude 2.5 ampere may be placed to the left of point R by 1m along x-axis then the magnetic induction at R will be zero.

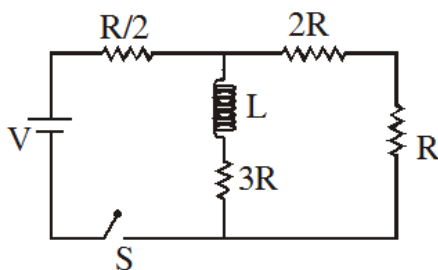
Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **Two (02)** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
 Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered)
 Negative Marks : -1 In all other cases

9. The circuit below has an ideal inductor and a switch as shown in figure. The switch can be closed for a long time and then can be opened at $t = 0$ or it can be closed at $t = 0$. List-I gives the different circuit elements and list-II gives the value.



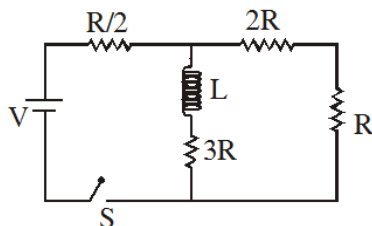
List-I		List-II	
(I)	Current in inductor	(P)	$\frac{6V}{7}$
(II)	Voltage across inductor	(Q)	$\frac{3V}{2}$
(III)	Current supplied by battery	(R)	$\frac{V}{4R}$
(IV)	Energy stored in the inductor	(S)	$\frac{V}{R}$
		(T)	Zero
		(U)	$\frac{LV^2}{32R^2}$

If the switch was closed at $t = 0$ then which of the following options are correct match of list I and II for that instant ?

- (A) $I \rightarrow S$ (B) $II \rightarrow P$ (C) $III \rightarrow T$ (D) $IV \rightarrow U$

Space for Rough Work

10. The circuit below has an ideal inductor and a switch as shown in figure. The switch can be closed for a long time and then can be opened at $t = 0$ or it can be closed at $t = 0$. List-I gives the different circuit elements and list-II gives the value.



List-I		List-II	
(I)	Current in inductor	(P)	$\frac{6V}{7}$
(II)	Voltage across inductor	(Q)	$\frac{3V}{2}$
(III)	Current supplied by battery	(R)	$\frac{V}{4R}$
(IV)	Energy stored in the inductor	(S)	$\frac{V}{R}$
		(T)	Zero
		(U)	$\frac{LV^2}{32R^2}$

If the switch remains closed for a long time and then is opened at $t = 0$ then which option in List-I is matched correctly with List-II at that instant ?

- (A) I $\rightarrow$ R (B) II $\rightarrow$ T (C) III $\rightarrow$ S (D) IV $\rightarrow$ T

Space for Rough Work

11. A string of length 'L' is tied between two rigid walls. The string is sustaining a standing wave and the tension in string is T. List-I gives the equation of standing wave and list-II gives different physical quantities for the standing waves in this string.

	List-I		List-II
(I)	$y = a \sin\left(\frac{\pi x}{L}\right) \cos \omega t$	(P)	Zero
(II)	$y = a \sin\left(\frac{\pi x}{L}\right) \sin \omega t$	(Q)	$\frac{a}{2}$
(III)	$y = a \sin\left(\frac{2\pi x}{L}\right) \sin \omega t$	(R)	$\frac{\pi^2 T a^2}{8L}$
(IV)	$y = a \sin\left(\frac{2\pi x}{L}\right) \cos \omega t$	(S)	a
		(T)	$\frac{L}{4}$
		(U)	$\frac{\pi^2 T a^2}{4L}$

The kinetic energy of the entire string at $t = 0$ is matched correctly in the following option.

- (A) I $\rightarrow$ R (B) I $\rightarrow$ U (C) II $\rightarrow$ U (D) II $\rightarrow$ P

Space for Rough Work

12. A string of length 'L' is tied between two rigid walls. The string is sustaining a standing wave and the tension in string is T. List-I gives the equation of standing wave and list-II gives different physical quantities for the standing waves in this string.

	List-I		List-II
(I)	$y = a \sin\left(\frac{\pi x}{L}\right) \cos \omega t$	(P)	Zero
(II)	$y = a \sin\left(\frac{\pi x}{L}\right) \sin \omega t$	(Q)	$\frac{a}{2}$
(III)	$y = a \sin\left(\frac{2\pi x}{L}\right) \sin \omega t$	(R)	$\frac{\pi^2 T a^2}{8L}$
(IV)	$y = a \sin\left(\frac{2\pi x}{L}\right) \cos \omega t$	(S)	a
		(T)	$\frac{L}{4}$
		(U)	$\frac{\pi^2 T a^2}{4L}$

The transverse displacement of the particle at a distance of $\frac{L}{12}$ from left end at time $t = 0$ is matched correctly in.

- (A) I $\rightarrow$ S (B) IV $\rightarrow$ Q (C) III $\rightarrow$ Q (D) IV $\rightarrow$ S

Space for Rough Work

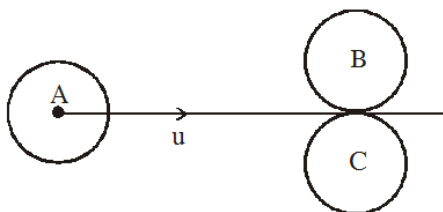
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

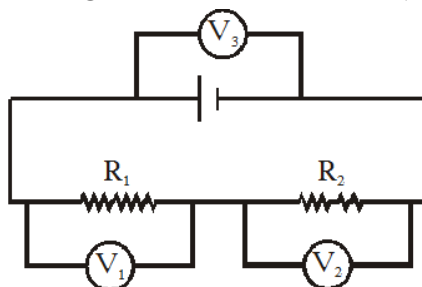
Zero Marks : 0 In all other cases.

1. An electron approaches a fixed proton from a large distance with a kinetic energy of 12.2 eV. The electron gets captured by the proton to form a hydrogen atom in an excited state and a photon of wavelength $\lambda_1 = 796\text{\AA}$ is emitted. Later, the hydrogen atom de-excites by emitting a single photon of wavelength λ_2 . Find λ_2 (in nm). [Take $h = 6.62 \times 10^{-34}$ Js, $c = 3 \times 10^8$ m/s]
2. On a billiard table two balls B and C are at rest touching each other. A third ball A, travelling with speed u , strikes the two balls elastically. Somehow A hits B first and within a fraction of a second hits ball C. You may assume that B and C are placed symmetrically with respect to the line of motion of A and that all the balls are identical. What angle (in degrees) does the final velocity of A make with its original direction of motion ?

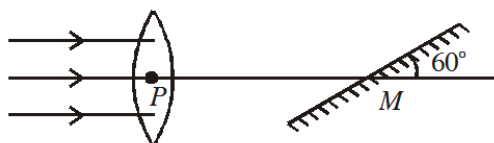


Space for Rough Work

3. In the circuit shown in the figure two resistor R_1 and R_2 have been connected in series to an ideal cell. When a voltmeter is connected across R_1 reading is $V_1 = 4.0$ volt and when the same voltmeter is connected across R_2 its reading is $V_2 = 6.0$ volt. The reading of the voltmeter when it is connected across the cell is $V_3 = 12.0$ volt. The actual voltage across R_1 in the circuit (in volt) is :

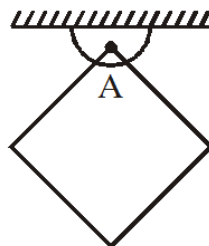


4. A horizontal parallel beam of light passes through a vertical convex lens of focal length f . The optical centre of the lens is P . A small plane mirror is placed at point M inclined at 60° to the axis of the lens. Distance $PM = \frac{f}{2}$. The mirror reflects the light passing through the lens and forms an image at point I . The distance PI is nf . Find n .



Space for Rough Work

5. A satellite of mass m is orbiting around the earth (mass M , radius R) in a circular orbit of radius $4R$. It starts losing energy slowly at constant rate $-\frac{dE}{dt} = \eta$ due to friction. The time t in which the satellite will spiral down to the surface of the earth is given by, $t = \frac{3GMm}{x\eta R}$. Find x .
6. A square plate of mass M and side length L is hinged at one of its vertex (A) and is free to rotate about it. The plate performs oscillations about a horizontal axis passing through A lying in the plane of the figure. The time period of small oscillations is given by $T = 2\pi \sqrt{\frac{nL}{\sqrt{2}g}}$. Find n .



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

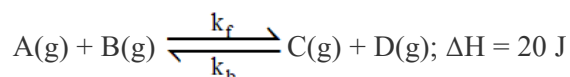
Negative Marks : -1 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

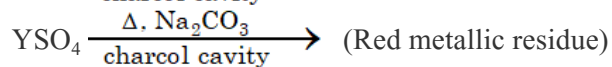
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1. Beryllium and aluminium exhibit many properties which are similar. Select the correct similar properties from the following.
(A) Exhibiting maximum covalency in compounds
(B) Forming polymeric hydrides
(C) Forming covalent halides
(D) Exhibiting amphoteric nature in their oxides
 2. Which of the following sets contain only addition polymers ?
(A) Polyethylene, polypropylene, terylene
(B) Polyethylene, PVC, orlon
(C) Buna-S, nylon, polybutadiene
(D) Bakelite, PVC, polyethylene
-

Space for Rough Work

3. Which of the following is/are correct for the reaction with equilibrium constant K ?



- (A) K_{eq} will increase on increasing temperature
 (B) k_f will increase on increasing temperature
 (C) k_b will increase on increasing temperature
 (D) k_b will decrease on increasing temperature
4. Two different metal sulphates XSO_4 & YSO_4 fused with Na_2CO_3 in charcol cavity, following observation is noted in residue :-

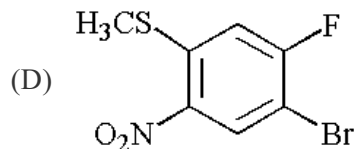
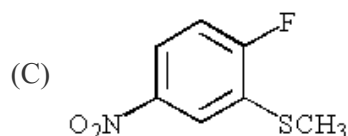
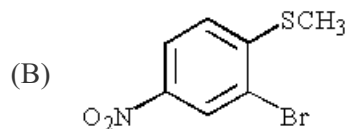
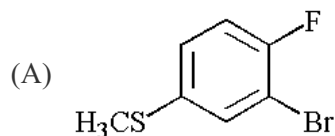
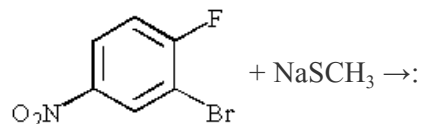


X^{+2} & Y^{+2} are respectively -

- (A) Pb^{+2} , Cu^{+2} (B) Zn^{+2} , Cu^{+2} (C) Zn^{+2} , Cd^{+2} (D) Zn^{+2} , Mg^{+2}

Space for Rough Work

5. Identify the principal organic product of the following reaction



6. Which of the following are correct statements

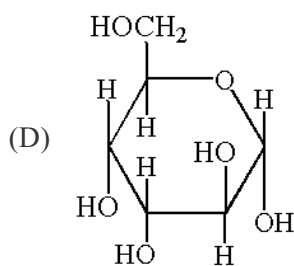
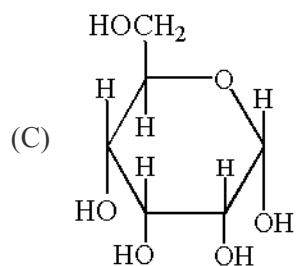
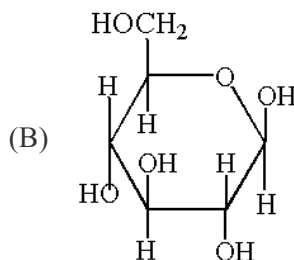
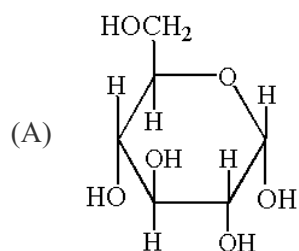
- (A) Spontaneous adsorption of gases on solid surface is an exothermic process as entropy decreases during adsorption.
- (B) Formation of micelles takes place when temperature is below Kraft Temperature (T_k) and concentration is above critical micelle concentration (CMC).
- (C) A colloid of Fe(OH)_3 is prepared by adding a little excess (required to completely precipitate Fe^{3+} ions as Fe(OH)_3) of NaOH in FeCl_3 solution. The particles of this sol will move towards cathode during electrophoresis.
- (D) According to Hardy-Schulze rules the coagulation (flocculating) value of Fe^{3+} ion will be more than Ba^{2+} and Na^+ .

Space for Rough Work

7. In which of the following both ions have S-S linkage ?

- (A) $\text{S}_2\text{O}_8^{2-}$, $\text{S}_4\text{O}_6^{2-}$ (B) $\text{S}_2\text{O}_6^{2-}$, $\text{S}_4\text{O}_6^{2-}$ (C) $\text{S}_2\text{O}_5^{2-}$, $\text{S}_2\text{O}_6^{2-}$ (D) $\text{S}_2\text{O}_3^{2-}$, $\text{S}_2\text{O}_5^{2-}$

8. Which of the following structures do not represents α D-glucopyranose ?



Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **Two (02)** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.

Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered)

Negative Marks : -1 In all other cases

9.

	List-I		List-II
(I)	$V(CO)_5$	(P)	metal metal bond is present
(II)	$Mn(CO)_6^{2+}$	(Q)	no metal metal bond is present
(III)	$Mn_2(CO)_{10}$	(R)	at least one bridge CO is present
(IV)	$[Co_2(CO)_8]$	(S)	no bridge CO is present
		(T)	metal-carbon bond order is greater than 1
		(U)	bond length of CO in complex is greater than free CO.

Which of the following is correct ?

(A) III $\rightarrow$ P,S,T,U

(B) I $\rightarrow$ T,U,R

(C) II $\rightarrow$ T,U,R

(D) IV $\rightarrow$ Q,R,T,U

Space for Rough Work

10.

	List-I		List-II
(I)	$\text{V}(\text{CO})_5$	(P)	metal metal bond is present
(II)	$\text{Mn}(\text{CO})_6^{2+}$	(Q)	no metal metal bond is present
(III)	$\text{Mn}_2(\text{CO})_{10}$	(R)	at least one bridge CO is present
(IV)	$[\text{Co}_2(\text{CO})_8]$	(S)	no bridge CO is present
		(T)	metal-carbon bond order is greater than 1
		(U)	bond length of CO in complex is greater than free CO.

Which of the following is incorrect ?

 (A) $\text{I} \rightarrow \text{Q}, \text{S}, \text{T}, \text{U}$

 (B) $\text{II} \rightarrow \text{Q}, \text{S}, \text{T}, \text{U}$

 (C) $\text{III} \rightarrow \text{P}, \text{R}, \text{T}, \text{U}$

 (D) $\text{IV} \rightarrow \text{P}, \text{R}, \text{T}, \text{U}$

Space for Rough Work

11.

	List-I		List-II
(I)	$\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{Cl} \xrightarrow[2.\text{H}_2\text{O}]{1.\text{CH}_3\text{MgBr}(\text{excess})}$	(P)	1° alcohol
(II)	$\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3 \xrightarrow[2.\text{H}_2\text{O}]{1.\text{CH}_3\text{MgBr}(\text{excess})}$	(Q)	2° alcohol
(III)	$\text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H} \xrightarrow[2.\text{H}_2\text{O}]{1.\text{CH}_3\text{MgBr}(\text{excess})}$	(R)	3° alcohol
(IV)	$\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{H} \xrightarrow[2.\text{H}_2\text{O}]{1.\text{CH}_3\text{MgBr}(\text{excess})}$	(S)	Positive iodoform test
		(T)	1 mole of grignard reagent is used per mole of reactant
		(U)	More than 1 mole of grignard reagent is used per mole of reactant

Which of the following is correct ?

(A) III → P,S,T

(B) I → P,S,T

(C) II → Q,U

(D) IV → P,SU

Space for Rough Work

12.

	List-I		List-II
(I)	$\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{Cl} \xrightarrow[2.\text{H}_2\text{O}]{1.\text{CH}_3\text{MgBr}(\text{excess})}$	(P)	1° alcohol
(II)	$\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3 \xrightarrow[2.\text{H}_2\text{O}]{1.\text{CH}_3\text{MgBr}(\text{excess})}$	(Q)	2° alcohol
(III)	$\text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H} \xrightarrow[2.\text{H}_2\text{O}]{1.\text{CH}_3\text{MgBr}(\text{excess})}$	(R)	3° alcohol
(IV)	$\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{H} \xrightarrow[2.\text{H}_2\text{O}]{1.\text{CH}_3\text{MgBr}(\text{excess})}$	(S)	Positive iodoform test
		(T)	1 mole of grignard reagent is used per mole of reactant
		(U)	More than 1 mole of grignard reagent is used per mole of reactant

Which of the following is incorrect ?

(A) I → R,S,U

(B) II → R,T

(C) III → P,S,T

(D) IV → Q,S,T

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

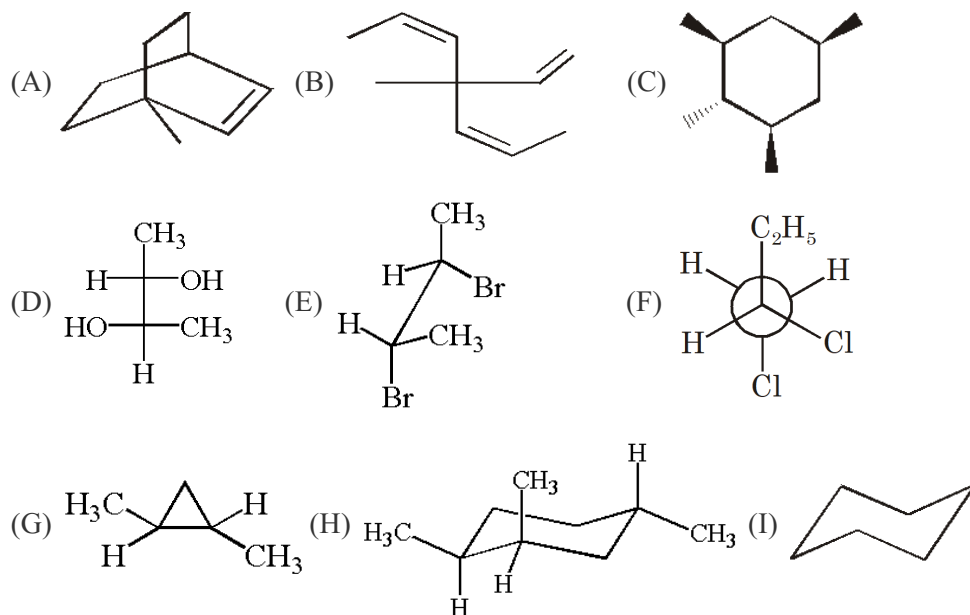
-
1. When 1 L of 1.3 M SnSO_4 is electrolysed for a period of 10 minutes using a current of 96.5 A and inert electrodes, the final concentration of Sn^{2+} , in the solution will be.
 $(E^\circ_{\text{Sn}^{2+}/\text{Sn}} = -0.14 \text{ V}, E^\circ_{\text{Sn}^{2+}/\text{Sn}^{4+}} = -0.13 \text{ V}, E^\circ_{\text{OH}^-/\text{O}_2} = -0.4 \text{ V}; (E^\circ_{\text{SO}_4^{2-}/\text{S}_2\text{O}_8^{2-}} = -2.0 \text{ V})$
 2. An athlete in a gymnasium room lifts a 50 kg mass through a vertical distance of 2.0 m; $g = 9.8 \text{ ms}^{-2}$. The mass is allowed to fall through 2.0 m distance while coupled to an electrical generator. The electrical generator produces an equal amount of electrical work. This electrical work is used to produce aluminum by Hall's process involving the change, Al_2O_3 (molten) + $3\text{C}(\text{s}) + 2\text{Al}(\text{l}) + 3\text{CO}(\text{g})$. The reaction require standard free energy change equal to 588 kJ. How many times (in Hundred) must the athlete lift the 50 kg mass and drop to couple the generator to produce sufficient Gibbs free energy to produce 27g Al ?
 3. What is the maximum number of layers of atoms in close packed planes that will lie within two imaginary parallel planes having a distance between them of $12\sqrt{\frac{2}{3}} R$ (where R is the radius of atom) ? (Consider the atoms to be within the parallel planes if their centres are on or within the two parallel planes).
-

Space for Rough Work

4. How many of the following are oxide ores ?

Bauxite, Cuprite, Cassiterite, Haematite, Magnetite, Chromite, Cryolite, galena, Limestone, Rutile, Pyrolusite.

5. Which of the following is/are chiral compounds?



6. Using van der Waals' equation of state, calculate pressure (in atm) developed by 132 g of CO_2 contained in a volume of 6.0 litre at 27°C .

($a = 3.6 \text{ atm L}^2 \text{ mol}^{-2}$, $b = 400 \text{ cm}^3 \text{ mol}^{-1}$, $R = 0.08 \text{ L-atm/K-mol}$)

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

-
1. The value of $[\vec{a} \times \vec{p} \vec{b} \times \vec{q} \vec{c} \times \vec{r}] + [\vec{a} \times \vec{q} \vec{b} \times \vec{r} \vec{c} \times \vec{p}] + [\vec{a} \times \vec{r} \vec{b} \times \vec{p} \vec{c} \times \vec{q}]$ is
 (A) independent of $\vec{a}, \vec{b}$ and $\vec{c}$ (B) independent of $\vec{p}, \vec{q}$ and $\vec{r}$
 (C) 0 (D) dependent of $\vec{a}, \vec{b}$ and $\vec{c}$
 2. In triangle ABC, I is the incentre of the triangle. If R and r be the circumradius and inradius of this triangle, then value of (AI) (BI) (CI) is $a^b R^c$, where
 (A) $a + b - c = 5$ (B) $a + bc = 6$ (C) $a + b = 5$ (D) $a + 2b - 3c = 5$
-

Space for Rough Work

3. If A and B are two independent events of an experiment such that $P(A) = 1/2$ and $P(B) = 1/5$, then
- (A) $P(A \cup B) = 3/5$ (B) $P(A/B) = 1/2$
(C) $P(A/A \cup B) = 5/6$ (D) $P(A \cap B/A' \cup B') = 0$
4. Let S_1 (with centre C_1 and radius $(r_1) = 2$ units) and S_2 (with centre C_2 and radius $(r_2) = 3$ units) be two fixed circles touching each other externally. Let S_3 (with centre C_3 and radius (r_3)) be a variable circle touching internally both S_1 and S_2 at points A and B respectively. The tangents to S_3 at A and B meet at T, and $TA = 4$. Which of the following is (are) TRUE?
- (A) r_3 is 6 units (B) r_3 is 8 units
(C) area of circle circumscribing ΔTAB is 10π (D) $C_3C_1 - C_3C_2 = 1$
-

Space for Rough Work

5. If y_1, y_2 are the solution of the differential equation $\frac{dy}{dx} + P(x)y = Q(x)$, then :
- (A) $y = y_1 + c(y_1 - y_2)$ is the general solution of equation, where C is arbitrary constant
- (B) $y = y_1 + c(y_1 + y_2)$ is the general solution of equation, where C is arbitrary constant
- (C) $\alpha y_1 + \beta y_2$ is a solution if $\alpha + \beta = 1$
- (D) $\alpha y_1 + \beta y_2$ is a solution if $\alpha - \beta = 1$
6. A book is having a few pages on which page numbers are written on both the sides of page. Somebody has torn one page out of it and now average of all the remaining page numbers is $\left(\frac{105}{4}\right)$. If total number of pages in the book is n and removed page numbers are k and k + 1, then which of the following is (are) CORRECT?
- (A) $n = 52$ (B) $n = 50$ (C) $k = 7$ (D) $k = 8$
7. Let A and B be two invertible matrices of order n such that $I - AB$ is invertible, then which of the following is (are) CORRECT?
- (A) $I - BA$ is singular (B) $I - BA$ is non-singular
- (C) $(I - BA)^{-1} = I + B(I - AB)^{-1}A$ (D) $(I - BA)^{-1} = I - B(I - AB)^{-1}A$
8. If the points of local extremum of $f(x) = x^3 - 3ax^2 + 3(a^2 - 1)x + 1$ lies between -2 & 4, then 'a' belongs to -
- (A) $(-2, 2)$ (B) $(-\infty, -1) \cup (3, \infty)$
- (C) $(-1, 3)$ (D) $(3, \infty)$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **Two (02)** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
 Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered)
 Negative Marks : -1 In all other cases

9. Match the following

List-I		List-II	
(I)	If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, $\vec{c} = \hat{i} + 2\hat{j} - \hat{k}$, then the value of $\begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix}$ is divisible by	(P)	6
(II)	If points P, Q and R have position vectors, $\vec{r}_1 = 3\hat{i} - 2\hat{j} - \hat{k}$, $\vec{r}_2 = \hat{i} + 3\hat{j} + 4\hat{k}$ and $\vec{r}_3 = 2\hat{i} + \hat{j} - 2\hat{k}$ respectively, relative to an origin O, then the value of distance of P from the plane OQR, divides.....	(Q)	28
(III)	A line with direction ratios (2, 1, 2) intersects the lines, $\vec{r} = -\hat{j} + \lambda(\hat{i} + \hat{j} + \hat{k})$ and $\vec{r} = -\hat{i} + \mu(2\hat{i} + \hat{j} + \hat{k})$ at A and B, then the value of length (AB) divides.....	(R)	16
(IV)	In a regular tetrahedron, the centres of the four faces are the vertices of a smaller tetrahedron. The ratio of the volume of the smaller tetrahedron to that larger is $\frac{m}{n}$, where m and n are relatively prime positive integers. The value of (m + n) is	(S)	3
		(T)	8
		(U)	10

Which of the following is the only correct combination ?

- (A) I $\rightarrow$ R,T (B) II $\rightarrow$ R,T (C) I $\rightarrow$ Q,T (D) II $\rightarrow$ P,T

Space for Rough Work

10. Match the following

List-I		List-II	
(I)	If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, $\vec{c} = \hat{i} + 2\hat{j} - \hat{k}$, then the value of $\begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix}$ is divisible by	(P)	6
(II)	If points P, Q and R have position vectors, $\vec{r}_1 = 3\hat{i} - 2\hat{j} - \hat{k}$, $\vec{r}_2 = \hat{i} + 3\hat{j} + 4\hat{k}$ and $\vec{r}_3 = 2\hat{i} + \hat{j} - 2\hat{k}$ respectively, relative to an origin O, then the value of distance of P from the plane OQR, divides.....	(Q)	28
(III)	A line with direction ratios (2, 1, 2) intersects the lines, $\vec{r} = -\hat{j} + \lambda(\hat{i} + \hat{j} + \hat{k})$ and $\vec{r} = -\hat{i} + \mu(2\hat{i} + \hat{j} + \hat{k})$ at A and B, then the value of length (AB) divides.....	(R)	16
(IV)	In a regular tetrahedron, the centres of the four faces are the vertices of a smaller tetrahedron. The ratio of the volume of the smaller tetrahedron to that larger is $\frac{m}{n}$, where m and n are relatively prime positive integers. The value of (m + n) is	(S)	3
		(T)	8
		(U)	10

Which of the following is the only correct combination ?

(A) III $\rightarrow$ R,T

(B) IV $\rightarrow$ T

(C) III $\rightarrow$ P,S

(D) IV $\rightarrow$ R

Space for Rough Work

11.

List-I		List-II	
(I)	If $\cos^{-1}\sqrt{x} + \cos^{-1}\sqrt{1-x} + \cos^{-1}\sqrt{1-y} = \frac{2\pi}{3}$, then $4(x + y)$ can be equal to	(P)	-1
(II)	$\lim_{z \rightarrow 0} \left[\left\{ \max \left((\sin^{-1}x + \cos^{-1}x)^2, \min(x^2 + 4x + 7) \right) \right\} \cdot \frac{\sin^{-1}z}{z} \right]$ is equal to (where $[.]$ denotes greatest integer function)	(Q)	1
(III)	If α is a solution of equation $ \sin x - \cos x = \sin 2x - 1$ in $[0, 2\pi]$, then $\frac{4\alpha}{\pi}$ can be equal to	(R)	3
(IV)	The integers in range of $f(x) = \begin{cases} \operatorname{sgn}(x) & x > 1 \\ x & x \leq 1 \end{cases}$ is/are	(S)	0
		(T)	5
		(U)	6

Which of the following is the only correct combination ?

(A) $I \rightarrow S$

(B) $II \rightarrow R$

(C) $I \rightarrow P$

(D) $II \rightarrow T$

Space for Rough Work

12.

List-I		List-II	
(I)	If $\cos^{-1} \sqrt{x} + \cos^{-1} \sqrt{1-x} + \cos^{-1} \sqrt{1-y} = \frac{2\pi}{3}$, then $4(x+y)$ can be equal to	(P)	-1
(II)	$\lim_{z \rightarrow 0} \left[\left\{ \max \left((\sin^{-1} x + \cos^{-1} x)^2, \min(x^2 + 4x + 7) \right) \right\} \cdot \frac{\sin^{-1} z}{z} \right]$ is equal to (where $[.]$ denotes greatest integer function)	(Q)	1
(III)	If α is a solution of equation $ \sin x - \cos x = \sin 2x - 1$ in $[0, 2\pi]$, then $\frac{4\alpha}{\pi}$ can be equal to	(R)	3
(IV)	The integers in range of $f(x) = \begin{cases} \operatorname{sgn}(x) & x > 1 \\ x & x \leq 1 \end{cases}$ is/are	(S)	0
		(T)	5
		(U)	6

Which of the following is the only correct combination ?

(A) $IV \rightarrow T$

(B) $IV \rightarrow R$

(C) $III \rightarrow P, R$

(D) $III \rightarrow Q, T$

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. If $f(x) = x^3 + ax^2 + bx + c = 0$ has three distinct integral roots and $(x^2 + 2x + 2)^3 + a(x^2 + 2x + 2)^2 + b(x^2 + 2x + 2) + c = 0$ has no real roots, then minimum value of a is _____.
2. If the line $y = mx + 2$ cuts the parabola $2y = x^2$ at points (x_1, y_1) and (x_2, y_2) ($x_1 < x_2$), then value of m for which $\int_{x_1}^{x_2} \left(mx + 2 - \frac{x^2}{2} \right) dx$ is minimum is equal to _____.
3. If $f(x) = x^3 + 3x^2 - 1$, then number of real solutions of $f(f(x)) = 0$, is _____.

Space for Rough Work

4. In a ΔABC , the vertex A is (1, 1) and its orthocenter is (2, 4). If the sides AB and BC are members of the family of straight lines $ax + by + c = 0$, where a, b, c are in A.P., then the coordinates of vertex C are (h, k). The value of $2h + 12k$ is equal to _____.
5. Let $f(x) = \begin{cases} \tan^{-1} x, & ; |x| \leq 1 \\ \frac{\pi}{4} \operatorname{sgn}(x) + \frac{x-1}{2}, & ; |x| > 1 \end{cases}$, (where $\operatorname{sgn}(x)$ denotes signum function)
Then the value of $f'_+(1)$ is equal to _____.
6. Let A, B, C be the set of complex numbers defined as $A = \{z: ||z + 2| - |z - 2|| = 2\}$,
 $B = \left\{ z : \arg\left(\frac{z-1}{z}\right) = \frac{\pi}{2} \right\}$ and $C = \{z: \arg(z-1) = \pi\}$, then $n(A \cap B \cap C)$ is _____.

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

Space for Rough Work